Course Title: Formal Methods in Software Engineering

Credit Hours: 3 (3,0)

Prerequisites: Discrete Structures

Course Learning Outcomes (CLOs):

- **CLO1: Describe the costs and benefits of formal methods.**
- **CLO2: Construct formal models of sequential software systems.**
- **CLO3: Implement sequential software systems based on formal models.**
- **CLO4: Verify attributes of formal models.**
- **CLO5: Demonstrate formal correctness of simple procedure.**

Primary Reference Material (Reading Material):

- **(Undergraduate Topics in Computer Science) O'Regan, Gerard - Concise Guide to Formal Methods _ Theory, Fundamentals and Industry Applications-Springer International Publishing (2017)**

Modern Formal Methods and Applications Edited by HOSSAM A. GABBAR Okayama University, Okayama, Japan

Additional Reference Materials (General Course References):

1. Guide to the Software Engineering Body of Knowledge v4.0 EDITOR Hironori Washizaki Waseda University, IEEE Computer Society 2024 President-Elect, 2025 President

Teaching Methodology: Lecturing, Written Assignments, Project, Report Writing

Course Assessment: Sessional 25% (Home Assignments, Quizzes, Project, Presentations)

Mid Term 35%

Final Exam 40%

32-Lecture Course Outline

Lecture	Topic	CLOs Covered	Reading Material (O'Regan)	Assessments & Activities
Week 1: Introduction to Software Engineering & Formal Methods				
1	Introduction to Software Engineering & Challenges	CLO1	Ch 1.1-1.3	Introduce Course, CLOs, Assessment.

2	**Software Reliability and Dependability**	**CLO1**	**Ch 2.1-2.2.2**	**Quiz 1 (Intro, SE Challenges)**
Week 2: Formal Methods Overview & Basic Math Foundations				
3	**Overview of Formal Methods: Definition, Benefits, Applications**	**CLO1**	**Ch 3.1-3.3**	**Assignment 1 (Benefits/Costs of FMs)**
4	**Criticisms & Usability of Formal Methods; Technology Transfer**	**CLO1**	**Ch 3.17, 3.18, 18.3, 18.4**	
5	**Foundational Mathematics: Sets & Set Theory**	**CLO2**	**Ch 4.1-4.2.2**	**Quiz 2 (Basic Math: Sets)**
6	**Foundational Mathematics: Relations & Functions**	**CLO2**	**Ch 4.3-4.5**	
Week 3: History & Fundamentals of Logic				
7	**Short History of Logic: Syllogistic, Paradoxes**	**CLO2**	**Ch 5.1-5.3**	**Assignment 2 (Set/Relation/Function Problems)**
8	**Short History of Logic: Stoic, Boole, Frege**	**CLO2**	**Ch 5.4-5.7**	
Week 4: Propositional Logic				
9	**Propositional Logic: Introduction, Truth Tables, Connectives**	**CLO2, CLO4**	**Ch 6.1-6.2.2**	**Quiz 3 (History of Logic, Boole's Logic)**
10	**Propositional Logic: Properties, Proof in Calculus**	**CLO5**	**Ch 6.2.3-6.2.8**	
Week 5: Predicate Logic				
11	**Predicate Logic: Introduction, Predicates, Quantifiers**	**CLO2, CLO4**	**Ch 6.3-6.3.3**	**Assignment 3 (Propositional Logic Proofs)**
12	**Predicate Logic: Interpretation, Valuation, Applications, Semantic Tableaux**	**CLO4, CLO5**	**Ch 6.3.4-6.3.5**	
Week 6: Advanced Topics in Logic				
13	**Advanced Logic: Fuzzy Logic, Temporal Logic**	**CLO2, CLO4**	**Ch 7.1-7.3**	**Quiz 4 (Predicate Logic)**

14	**Advanced Logic: Intuitionistic Logic, Undefined Values (LPF, Parnas, Dijkstra)**	**CLO2, CLO4**	**Ch 7.4-7.5**	
Week 7: Z Formal Specification Language I				
15	**Z Formal Specification Language: Introduction, Sets, Relations**	**CLO2**	**Ch 8.1-8.3**	**Assignment 4 (Advanced Logic/Undefinedness)**
16	**Z: Functions, Sequences, Bags**	**CLO2**	**Ch 8.4-8.6**	
Week 8: Z Formal Specification Language II & Mid Term Exam				
17	**Z: Schemas and Schema Composition**	**CLO2**	**Ch 8.7**	**Sessional Exam 1 (Lectures 1-16)**
18	**Z: Reification, Decomposition, and Proof**	**CLO3, CLO5**	**Ch 8.8-8.9**	
Week 9: Vienna Development Method (VDM) I				
19	**VDM: Introduction, Sets, Sequences**	**CLO2**	**Ch 9.1-9.3**	**Introduce Project Scope/Topics.**
20	**VDM: Maps, Logic of Partial Functions in VDM**	**CLO2, CLO4**	**Ch 9.4-9.5**	**Quiz 5 (Z Specification)**
Week 10: Vienna Development Method (VDM) II				
21	**VDM: Data Types, Data Invariants, Specification, Refinement**	**CLO3, CLO5**	**Ch 9.6-9.8**	**Assignment 5 (VDM Specification/Refinement)**
22	**Irish School of VDM (VDM♠): Distinguishing Features, Structures, Refinement**	**CLO2, CLO3, CLO5**	**Ch 10.1-10.9**	
Week 11: Dijkstra, Hoare, Parnas Contributions				
23	**Dijkstra: Calculus of Weakest Preconditions (WP)**	**CLO5**	**Ch 12.1-12.2**	**Project Check-in 1 (Initial Model)**
24	**Hoare: Axiomatic Semantics of Programming Languages**	**CLO5**	**Ch 12.3**	
Week 12: Other Modelling & Verification Techniques				

25	**Parnas: Classical Engineering Approach, Tabular Expressions**	**CLO2**	**Ch 12.4**	**Quiz 6 (Dijkstra, Hoare, Parnas)**
26	**Automata Theory: Finite State Machines (FSM)**	**CLO2, CLO4**	**Ch 13.1-13.2**	
Week 13: Advanced Automata & Model Checking				
27	**Automata Theory: Pushdown Automata, Turing Machines**	**CLO2, CLO4**	**Ch 13.3-13.4**	**Assignment 6 (Automata/Tabular Expressions)**
28	**Model Checking: Introduction, Process, Linear Temporal Logic (LTL)**	**CLO4**	**Ch 14.1-14.3**	
Week 14: Model Checking & Theorem Proving				
29	**Model Checking: Computational Tree Logic (CTL), Tools, Applications**	**CLO4**	**Ch 14.4-14.6**	**Project Check-in 2 (Verification/Refinement Status)**
30	**The Nature of Proof and Theorem Proving, Automated & Interactive Provers**	**CLO5**	**Ch 15.1-15.4**	**Quiz 7 (Model Checking)**
Week 15: Tools & Practical Aspects				
31	**Logic and AI. Industrial Tools for Formal Methods**	**CLO1, CLO2**	**Ch 7.6, Ch 17**	
32	**Technology Transfer, Usability, and The Future of Formal Methods**	**CLO1**	**Ch 18.1-18.4, Ch 19**	**Project Presentations & Report Submission. Final Exam Review**